

## 20.3 Symmetric Gauge Functions

The last two sections turned a majorization into an inequality between sums of a convex function. This section turns it into an inequality between **norms**. The norms that work are the ones blind to exactly the information a majorization discards — the order of the entries and their signs — and Section 4 will show that these are precisely the norms a matrix inherits from its singular values.

Throughout, the field is  $\mathbb{R}$  and the space is  $\mathbb{R}^n$  with the dot product  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ ,  $n \geq 1$ . Majorization  $\prec$  and weak majorization  $\prec_w$  are Chapter 16 §08's (Definition 16.8.1), and  $\mathbf{x}^\downarrow$  is the decreasing rearrangement. A **permutation matrix**  $\mathbf{P}_\sigma$  is Definition 2.6.2;  $\mathbf{P}_\sigma \mathbf{x}$  rearranges the entries of  $\mathbf{x}$ . We write  $|\mathbf{x}| = (|x_1|, \dots, |x_n|)$ , and  $\mathbf{x} \leq \mathbf{y}$  means  $x_i \leq y_i$  for **every**  $i$  — the entrywise order of Chapter 17 §05, never the Loewner order. From §01 we use Corollary 20.1.4, which rests on Theorem 20.1.3; the exercises also use §02's Theorem 20.2.1.

### 20.3.1 What a norm must forget

Chapter 16 §06 measured a Hermitian matrix by its Ky Fan partial sum  $s_k(\mathbf{A}) = \lambda_1(\mathbf{A}) + \dots + \lambda_k(\mathbf{A})$ , and Exercise (C2: Partial sums that behave like a norm) asked whether  $s_k$  is a norm. On the positive semidefinite matrices it behaves like one; on all Hermitian matrices it fails, because  $s_1(\text{diag}(0, -1)) = 0$  with  $\text{diag}(0, -1) \neq \mathbf{0}$ . The exercise ended by naming the repair: apply  $s_k$  to the **singular** values instead of the eigenvalues. Singular values are non-negative, so nothing can cancel, and the failure disappears.

That repair suggests the shape of the whole chapter's answer. A measurement of a matrix built from its singular values is a function of a list of  $n$  non-negative numbers. Two things about that list are accidents of bookkeeping. Its **order** is one: the singular values of a matrix form a multiset, and Chapter 12 chose to write them decreasingly. Its **signs** are the other: a formula that is to extend from non-negative lists to all of  $\mathbb{R}^n$ , and be a norm there, cannot notice which entries are negative, since  $\mathbf{x}$  and the vector obtained by flipping one sign have the same absolute values and must be measured alike.

*A symmetric gauge function is a norm that cannot tell the entries of a vector apart by where they sit or which way they point.*

#### Definition 20.3.1 — Symmetric Gauge Function

A **symmetric gauge function** on  $\mathbb{R}^n$  is a norm  $\Phi: \mathbb{R}^n \rightarrow \mathbb{R}$  (Definition 15.1.1) such that

- (G1)  $\Phi(\mathbf{P}_\sigma \mathbf{x}) = \Phi(\mathbf{x})$  for **every** permutation matrix  $\mathbf{P}_\sigma$  and every  $\mathbf{x} \in \mathbb{R}^n$  (**permutation invariance**);
- (G2)  $\Phi(\varepsilon_1 x_1, \dots, \varepsilon_n x_n) = \Phi(\mathbf{x})$  for **all** signs  $\varepsilon_i \in \{+1, -1\}$  and every  $\mathbf{x} \in \mathbb{R}^n$  (**sign invariance**).

In words: (G1) says that shuffling the entries does not change the measurement, and (G2) says that flipping any of their signs does not either. The word **every** does the work in both, and the two clauses are genuinely separate: below are a norm with (G1) and not (G2), and one with (G2) and not (G1). Together they say that  $\Phi(\mathbf{x})$  depends only on the multiset  $\{|x_1|, \dots, |x_n|\}$ ,

that is,

$$\Phi(\mathbf{x}) = \Phi(|\mathbf{x}|^\downarrow) \quad (\mathbf{x} \in \mathbb{R}^n), \quad (20.3.1)$$

since  $|\mathbf{x}|$  is obtained from  $\mathbf{x}$  by (G2) and  $|\mathbf{x}|^\downarrow$  from  $|\mathbf{x}|$  by (G1). Note also that (N2) of Definition 15.1.1 already gives  $\Phi(-\mathbf{x}) = \Phi(\mathbf{x})$ ; (G2) is much stronger, because it flips the signs **one entry at a time**.

*Warning.* The word “gauge” is used here in a different sense from Chapter 17 §04. There, Definition 17.4.1 attached to a convex set  $K$  the function  $p_K(\mathbf{x}) = \inf\{t > 0 : \mathbf{x} \in tK\}$ , which measures a vector by inflating  $K$ . That object is not this one. The name “symmetric gauge function” is standard and traditional for Definition 20.3.1, and the two never appear in the same argument in this book; where confusion is possible, this chapter writes  $\Phi$  and never  $p_K$ .

The first examples are the norms already at hand.

**Example. (The  $p$ -norms, the Ky Fan gauges, and size one)**

Check each against Definition 20.3.1.

- (a)  $\|\cdot\|_p$  for  $1 \leq p \leq \infty$ .
- (b) For  $1 \leq k \leq n$ , the function  $\Phi_k(\mathbf{x}) = \sum_{i \leq k} |x|_i^\downarrow$ , the sum of the  $k$  largest absolute values.
- (c) On  $\mathbb{R}^1$ , the function  $\Phi(x) = c|x|$  for a constant  $c > 0$ .

*Solution.*

- (a) These are norms: Example (The 1-, 2- and  $\infty$ -norms on  $F^n$ ) for  $p = 1, 2, \infty$ , and Corollary 17.11.5 supplies the triangle inequality for the rest. The formulas  $(\sum_i |x_i|^p)^{1/p}$  and  $\max_i |x_i|$  are built from the absolute values alone and are unchanged by reordering the sum or the maximum, so (G1) and (G2) hold.
- (b) Rewrite the definition as a maximum:

$$\Phi_k(\mathbf{x}) = \max \left\{ \sum_{i \in S} |x_i| : S \subseteq \{1, \dots, n\}, |S| = k \right\},$$

since the largest such sum is the one over the indices of the  $k$  largest absolute values. Each function  $\mathbf{x} \mapsto \sum_{i \in S} |x_i|$  is non-negative, absolutely homogeneous and subadditive, so the maximum of finitely many of them is too, which gives (N2) and (N3). For (N1), every singleton is contained in some  $S$ , so  $\Phi_k(\mathbf{x}) \geq \max_i |x_i|$ , and  $\Phi_k(\mathbf{x}) = 0$  forces  $\mathbf{x} = \mathbf{0}$ . Both (G1) and (G2) are visible in the displayed formula. At the two ends,  $\Phi_1 = \|\cdot\|_\infty$  and  $\Phi_n = \|\cdot\|_1$ .

- (c) A norm on  $\mathbb{R}^1$  by inspection. There is only one permutation and  $\Phi(-x) = \Phi(x)$ , so (G1) and (G2) are automatic: on  $\mathbb{R}^1$  **every** norm is a symmetric gauge function. The degenerate case is worth keeping in mind, because it shows that the two clauses carry no information until  $n \geq 2$ .

The functions in (b) will be used constantly, so they get a name of their own.

**Definition 20.3.2 — Ky Fan Gauge**

For  $1 \leq k \leq n$ , the  $k$ -th **Ky Fan gauge** on  $\mathbb{R}^n$  is

$$\Phi_k(\mathbf{x}) = \sum_{i=1}^k |x_i|^\downarrow,$$

the sum of the  $k$  largest absolute values of the entries of  $\mathbf{x}$ .

**Non-examples by minimal change.** Each drops exactly one clause.

- A norm with (G1) and not (G2). On  $\mathbb{R}^2$  let

$$\Phi(\mathbf{x}) = \max(|x_1 + x_2|, |x_1|, |x_2|).$$

Each of the three expressions is non-negative, absolutely homogeneous and subadditive, so their maximum is too; and  $\Phi(\mathbf{x}) = 0$  forces  $x_1 = x_2 = 0$ , so  $\Phi$  is a norm. Swapping  $x_1$  and  $x_2$  permutes the three expressions, so (G1) holds. But  $\Phi(1, 1) = 2$  while  $\Phi(1, -1) = 1$ : (G2) fails.

- A norm with (G2) and not (G1). On  $\mathbb{R}^2$  let  $\Phi(\mathbf{x}) = |x_1| + 2|x_2|$ . This is a norm, and it is built from absolute values, so (G2) holds. But  $\Phi(1, 0) = 1$  and  $\Phi(0, 1) = 2$ : (G1) fails.

**Why this definition.** The two clauses are exactly what a norm needs in order to be computable from an unordered list of non-negative numbers, and nothing more is demanded — in particular, no relation to any other norm, and no smoothness. Dropping (G1) allows a norm to prefer one coordinate to another, which no measurement of singular values can do, since relabeling the singular values of a matrix is not an operation on the matrix at all. Dropping (G2) allows a norm to see cancellation between coordinates, which is fatal: Section 7 needs  $\Phi$  evaluated at differences of eigenvalue lists and of singular value lists, whose entries have both signs (Theorem 20.7.2 and Theorem 20.7.4).

*Quick check.* Is  $\Phi(\mathbf{x}) = \max_i x_i$  (no absolute values) a symmetric gauge function on  $\mathbb{R}^n$  for  $n \geq 2$ ? Which clause fails first?

*Solution.* It is not, and it fails before (G1) or (G2) are reached: it is not a norm. Absolute homogeneity (N2) fails at  $c = -1$ , since  $\max_i (-x_i) = -\min_i x_i$ , which for  $\mathbf{x} = (1, 0)$  is 0 while  $\max_i x_i = 1$ . Positive definiteness (N1) fails too:  $\Phi(-1, -1) = -1 < 0$ . Restricted to vectors with non-negative entries the formula is perfectly reasonable, and it equals  $\|\mathbf{x}\|_\infty$  there; the definition insists on a norm on all of  $\mathbb{R}^n$  precisely so that this kind of one-sided formula is excluded.

## 20.3.2 A symmetric gauge is monotone

The first payoff is that a symmetric gauge respects the crudest comparison there is: if every entry of  $\mathbf{x}$  is smaller in absolute value than the matching entry of  $\mathbf{y}$ , then  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$ . This is not automatic for a norm. The function  $N(\mathbf{x}) = |x_1 - x_2| + |x_1|$  is a norm on  $\mathbb{R}^2$ , and  $N(1, 0) = 2$  is **larger** than  $N(1, 1) = 1$  although  $(1, 0)$  is dominated entrywise by  $(1, 1)$ . Sign invariance is what rules this out.

### Proposition 20.3.1 — Symmetric Gauges Are Monotone

Let  $\Phi$  be a symmetric gauge function on  $\mathbb{R}^n$ , and let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  satisfy  $|x_i| \leq |y_i|$  for every  $i$ . Then  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$ .

*Idea.* Change one coordinate at a time. Shrinking a single entry towards 0 is taking an average of two vectors: the vector itself and the one with that entry's sign flipped. Sign invariance says those two have the same  $\Phi$ , and the triangle inequality says the average has no larger one.

*Proof. Step 1: one coordinate.* Suppose  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$  agree in every coordinate except the  $j$ -th, where  $|u_j| \leq |v_j|$ . If  $v_j = 0$  then  $u_j = 0$  and  $\mathbf{u} = \mathbf{v}$ . Otherwise put

$$\theta = \frac{1}{2} \left( 1 + \frac{u_j}{v_j} \right),$$

which lies in  $[0, 1]$  because  $|u_j/v_j| \leq 1$ . Let  $\mathbf{v}'$  be  $\mathbf{v}$  with its  $j$ -th coordinate negated. Then  $\theta v_j + (1 - \theta)(-v_j) = u_j$ , while in every other coordinate  $\mathbf{v}$  and  $\mathbf{v}'$  agree with  $\mathbf{u}$ , so

$$\mathbf{u} = \theta \mathbf{v} + (1 - \theta) \mathbf{v}'.$$

By (N3) and (N2) of Definition 15.1.1,  $\Phi(\mathbf{u}) \leq \theta \Phi(\mathbf{v}) + (1 - \theta) \Phi(\mathbf{v}')$ , and  $\Phi(\mathbf{v}') = \Phi(\mathbf{v})$  by (G2). Hence  $\Phi(\mathbf{u}) \leq \Phi(\mathbf{v})$ .

**Step 2: all coordinates.** Put  $\mathbf{z}^{(0)} = \mathbf{y}$  and, for  $1 \leq j \leq n$ ,

$$\mathbf{z}^{(j)} = (x_1, \dots, x_j, y_{j+1}, \dots, y_n),$$

so that  $\mathbf{z}^{(n)} = \mathbf{x}$ . Consecutive vectors  $\mathbf{z}^{(j-1)}$  and  $\mathbf{z}^{(j)}$  agree except in coordinate  $j$ , where  $|x_j| \leq |y_j|$ . By Step 1,  $\Phi(\mathbf{z}^{(j)}) \leq \Phi(\mathbf{z}^{(j-1)})$  for each  $j$ . Chaining the  $n$  inequalities gives  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$ , as claimed. ■

Only (G2) was used. Permutation invariance is needed for the next step, and there it does all the work.

*Quick check.* Deduce from Proposition 20.3.1 that  $\Phi(\mathbf{x}) \geq |x_j| \Phi(\mathbf{e}_j)$  for every  $j$ . What does the same proposition give as an upper bound, using  $\mathbf{x} = \sum_j x_j \mathbf{e}_j$ ?

*Solution.* The vector  $x_j \mathbf{e}_j$  has  $j$ -th entry  $x_j$  and all others 0, so its entries are dominated in absolute value by those of  $\mathbf{x}$ ; Proposition 20.3.1 gives  $|x_j| \Phi(\mathbf{e}_j) = \Phi(x_j \mathbf{e}_j) \leq \Phi(\mathbf{x})$ . For the upper bound the triangle inequality suffices:  $\Phi(\mathbf{x}) \leq \sum_j |x_j| \Phi(\mathbf{e}_j)$ . By (G1) all the numbers  $\Phi(\mathbf{e}_j)$  are equal, to  $\Phi(\mathbf{e}_1)$  say, so

$$\Phi(\mathbf{e}_1) \|\mathbf{x}\|_\infty \leq \Phi(\mathbf{x}) \leq \Phi(\mathbf{e}_1) \|\mathbf{x}\|_1.$$

Every symmetric gauge is trapped between multiples of  $\|\cdot\|_\infty$  and  $\|\cdot\|_1$ , with the same constant at both ends. Section 4 turns this into the statement that every **normalized** unitarily invariant norm lies between the spectral norm and the trace norm.

Both bounds are cited later, so they are recorded here.

### Corollary 20.3.2 — A Symmetric Gauge Is Trapped Between Two Norms

Let  $\Phi$  be a symmetric gauge function on  $\mathbb{R}^n$ . Then  $\Phi(\mathbf{e}_j) = \Phi(\mathbf{e}_1)$  for every  $j$ , and

$$\Phi(\mathbf{e}_1)\|\mathbf{x}\|_\infty \leq \Phi(\mathbf{x}) \leq \Phi(\mathbf{e}_1)\|\mathbf{x}\|_1 \quad (\mathbf{x} \in \mathbb{R}^n).$$

*Proof.* Fix  $j$ . Some permutation matrix carries  $\mathbf{e}_j$  to  $\mathbf{e}_1$ , so  $\Phi(\mathbf{e}_j) = \Phi(\mathbf{e}_1)$  by (G1). The entries of  $x_j\mathbf{e}_j$  are dominated in absolute value by those of  $\mathbf{x}$ , so Proposition 20.3.1 gives

$$|x_j| \Phi(\mathbf{e}_1) = \Phi(x_j\mathbf{e}_j) \leq \Phi(\mathbf{x}),$$

where the equality is (N2) of Definition 15.1.1; taking the largest  $|x_j|$  gives the left-hand inequality. For the right-hand one,  $\mathbf{x} = \sum_j x_j\mathbf{e}_j$ , so (N3) and (N2) give  $\Phi(\mathbf{x}) \leq \sum_j |x_j| \Phi(\mathbf{e}_j) = \Phi(\mathbf{e}_1)\|\mathbf{x}\|_1$ . This proves the corollary. ■

## 20.3.3 From weak majorization to a dominating vector

Now for the bridge. A majorization  $\mathbf{x} \prec \mathbf{y}$  will be handled by Corollary 20.1.4, which writes  $\mathbf{x}$  as an average of rearrangements of  $\mathbf{y}$ ; a norm cannot grow under an average. Weak majorization loses the equality of totals, so  $\mathbf{x}$  need not be such an average —  $(1, 0) \prec_w (2, 0)$ , but  $(1, 0)$  is no average of  $(2, 0)$  and  $(0, 2)$ . What survives is that  $\mathbf{x}$  is dominated **entrywise** by something that is such an average, and that is exactly the hypothesis Proposition 20.3.1 wants.

### Lemma 20.3.3 — Weak Majorization Is Domination plus Majorization

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Then  $\mathbf{x} \prec_w \mathbf{y}$  if and only if there is a vector  $\mathbf{z} \in \mathbb{R}^n$  with

$$\mathbf{x} \leq \mathbf{z} \text{ entrywise,} \quad \mathbf{z} \prec \mathbf{y}.$$

If  $\mathbf{x}$  has **non-negative** entries, then  $\mathbf{z}$  may be taken with non-negative entries as well.

*Idea.*  $(\Leftarrow)$  is two easy monotonicity steps. For  $(\Rightarrow)$ , sort both vectors and think of the deficit  $\sum_i y_i - \sum_i x_i$  as water to be poured into  $\mathbf{x}$ . Pouring it into the **top** entries would break the partial-sum inequalities immediately, so pour it into the bottom: raise every entry of  $\mathbf{x}$  that lies below a level  $c$  up to  $c$ , and choose  $c$  so that the totals match. Choosing  $c$  needs no analysis: on each interval between consecutive entries of  $\mathbf{x}$  the total is an affine function of  $c$ , so the right  $c$  can be written down.

*Proof.*  $(\Leftarrow)$  Suppose  $\mathbf{x} \leq \mathbf{z}$  entrywise and  $\mathbf{z} \prec \mathbf{y}$ . Fix  $k$  and let  $I$  be a set of  $k$  indices

carrying the  $k$  largest entries of  $\mathbf{x}$ , so that  $\sum_{i \leq k} x_i^\downarrow = \sum_{i \in I} x_i$ . Then

$$\sum_{i \leq k} x_i^\downarrow = \sum_{i \in I} x_i \leq \sum_{i \in I} z_i \leq \sum_{i \leq k} z_i^\downarrow,$$

the last step because the sum of **any**  $k$  entries of  $\mathbf{z}$  is at most the sum of its  $k$  largest. Hence  $\mathbf{x} \prec_w \mathbf{z}$ , and  $\mathbf{z} \prec \mathbf{y}$  gives  $\sum_{i \leq k} z_i^\downarrow \leq \sum_{i \leq k} y_i^\downarrow$  for  $k < n$  and equality of totals at  $k = n$ ; in both cases  $\sum_{i \leq k} z_i^\downarrow \leq \sum_{i \leq k} y_i^\downarrow$ . Combining,  $\mathbf{x} \prec_w \mathbf{y}$ .

( $\Rightarrow$ ) Suppose  $\mathbf{x} \prec_w \mathbf{y}$ . Majorization and weak majorization depend only on the decreasing rearrangements (Definition 16.8.1 and the remark after it), and the entrywise relation  $\mathbf{x} \leq \mathbf{z}$  is preserved by applying one permutation to both sides. So it is enough to construct  $\mathbf{z}$  when  $\mathbf{x} = \mathbf{x}^\downarrow$  and  $\mathbf{y} = \mathbf{y}^\downarrow$ : for general  $\mathbf{x}$ , choose  $\mathbf{P}_\sigma$  with  $\mathbf{P}_\sigma \mathbf{x} = \mathbf{x}^\downarrow$ , build  $\mathbf{z}'$  for the pair  $(\mathbf{x}^\downarrow, \mathbf{y}^\downarrow)$ , and take  $\mathbf{z} = \mathbf{P}_\sigma^{-1} \mathbf{z}'$ .

So let  $x_1 \geq \cdots \geq x_n$  and  $y_1 \geq \cdots \geq y_n$  with  $\sum_{i \leq k} x_i \leq \sum_{i \leq k} y_i$  for every  $k$ . Write  $S = \sum_{i=1}^n y_i$  and, for  $0 \leq k \leq n-1$ ,

$$c_k = \frac{S - (x_1 + \cdots + x_k)}{n - k},$$

the level at which the last  $n - k$  entries would have to sit for the total to be  $S$ . At  $k = n - 1$  we have  $c_{n-1} = S - \sum_{i \leq n-1} x_i \geq \sum_{i \leq n} x_i - \sum_{i \leq n-1} x_i = x_n$ , using  $\mathbf{x} \prec_w \mathbf{y}$  at  $k = n$ . So the set of indices  $k \in \{0, \dots, n-1\}$  with  $c_k \geq x_{k+1}$  is non-empty; let  $j$  be its **smallest** element and put  $c = c_j$ . Define

$$\mathbf{z} = (x_1, \dots, x_j, \underbrace{c, \dots, c}_{n-j}).$$

*The vector  $\mathbf{z}$  dominates  $\mathbf{x}$ .* For  $i \leq j$  the entries agree. For  $i > j$ ,  $z_i = c \geq x_{j+1} \geq x_i$ , since  $\mathbf{x}$  is decreasing and  $c \geq x_{j+1}$  by the choice of  $j$ . If in addition  $\mathbf{x} \geq \mathbf{0}$ , then  $c \geq x_{j+1} \geq 0$  and every entry of  $\mathbf{z}$  is non-negative, which is the last claim of the lemma.

*The vector  $\mathbf{z}$  is decreasing.* Only the step from  $x_j$  to  $c$  is in question, so assume  $j \geq 1$ . By the minimality of  $j$  we have  $c_{j-1} < x_j$ . Multiplying out the two definitions,

$$(n - j + 1)c_{j-1} = S - \sum_{i \leq j-1} x_i = (n - j)c_j + x_j,$$

so  $(n - j)c = (n - j + 1)c_{j-1} - x_j < (n - j + 1)x_j - x_j = (n - j)x_j$ , and  $n - j \geq 1$  gives  $c < x_j$ . Hence  $\mathbf{z} = \mathbf{z}^\downarrow$ .

*The vector  $\mathbf{z}$  is majorized by  $\mathbf{y}$ .* Its total is  $\sum_{i \leq j} x_i + (n - j)c = S$  by the definition of  $c_j$ , which is (M2). For (M1) with  $k \leq j$ , the first  $k$  entries of  $\mathbf{z}$  are those of  $\mathbf{x}$ , so  $\sum_{i \leq k} z_i = \sum_{i \leq k} x_i \leq \sum_{i \leq k} y_i$ . For  $k > j$ , subtract from the totals: since  $\sum_i z_i = \sum_i y_i = S$ ,

$$\sum_{i \leq k} z_i - \sum_{i \leq k} y_i = \sum_{i > k} y_i - \sum_{i > k} z_i = \sum_{i > k} y_i - (n - k)c,$$

so it suffices to show  $\sum_{i>k} y_i \leq (n-k)c$ . First, at  $k = j$  this holds: from  $(n-j)c = S - \sum_{i \leq j} x_i$  and  $\sum_{i \leq j} x_i \leq \sum_{i \leq j} y_i$ ,

$$(n-j)c \geq S - \sum_{i \leq j} y_i = \sum_{i > j} y_i.$$

So  $c$  is at least the average of the  $n-j$  smallest entries of  $\mathbf{y}$ . Now let  $k > j$ . Each of the entries  $y_{j+1}, \dots, y_k$  is at least each of  $y_{k+1}, \dots, y_n$ , because  $\mathbf{y}$  is decreasing; hence each of them is at least the average  $A$  of  $y_{k+1}, \dots, y_n$ , and therefore

$$\sum_{i>j} y_i = \sum_{i=j+1}^k y_i + (n-k)A \geq (k-j)A + (n-k)A = (n-j)A.$$

Combining,  $(n-j)c \geq \sum_{i>j} y_i \geq (n-j)A$ , so  $c \geq A$  and  $(n-k)c \geq (n-k)A = \sum_{i>k} y_i$ , as required. This proves  $\mathbf{z} \prec \mathbf{y}$  and the lemma. ■

The construction is short enough to run by hand.

#### Example. (Filling from the bottom)

Let  $\mathbf{x} = (4, 1, 1)$  and  $\mathbf{y} = (5, 3, 2)$ . Verify  $\mathbf{x} \prec_w \mathbf{y}$  and produce a  $\mathbf{z}$  with  $\mathbf{x} \leq \mathbf{z}$  and  $\mathbf{z} \prec \mathbf{y}$ .

*Solution.* Both vectors are already decreasing. The running totals are 4, 5, 6 for  $\mathbf{x}$  and 5, 8, 10 for  $\mathbf{y}$ , and  $4 \leq 5$ ,  $5 \leq 8$ ,  $6 \leq 10$ , so  $\mathbf{x} \prec_w \mathbf{y}$ . The totals differ, so  $\mathbf{x} \not\prec \mathbf{y}$ .

Here  $S = 10$  and  $n = 3$ . The levels are  $c_0 = 10/3$  and  $c_1 = (10-4)/2 = 3$ . Since  $c_0 = 10/3 < 4 = x_1$ , the index  $k = 0$  fails; since  $c_1 = 3 \geq 1 = x_2$ , the smallest admissible index is  $j = 1$ , and  $c = 3$ . Hence

$$\mathbf{z} = (4, 3, 3).$$

Check:  $(4, 1, 1) \leq (4, 3, 3)$  entrywise;  $\mathbf{z}$  is decreasing; its running totals 4, 7, 10 satisfy  $4 \leq 5$ ,  $7 \leq 8$ ,  $10 = 10$ , so  $\mathbf{z} \prec \mathbf{y}$ . The deficit of 4 between the totals was poured entirely into the two bottom entries, which is what keeps the top partial sums under control.

## 20.3.4 The inequality

Everything is now in place.

### Theorem 20.3.4 — Symmetric Gauges Respect Majorization

Let  $\Phi$  be a symmetric gauge function on  $\mathbb{R}^n$ .

- (a) If  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and  $\mathbf{x} \prec \mathbf{y}$ , then  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$ .
- (b) If  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  have **non-negative** entries and  $\mathbf{x} \prec_w \mathbf{y}$ , then  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$ .

*Idea.* For (a), §01 turns  $\mathbf{x} \prec \mathbf{y}$  into “ $\mathbf{x}$  is an average of rearrangements of  $\mathbf{y}$ ”, and a norm never exceeds an average of the values it takes on the points averaged — those values all

being  $\Phi(\mathbf{y})$ , by permutation invariance. For (b), [Lemma 20.3.3](#) inserts a middle vector  $\mathbf{z}$ : from  $\mathbf{x}$  to  $\mathbf{z}$  use monotonicity, from  $\mathbf{z}$  to  $\mathbf{y}$  use (a).

*Proof.*

- (a) By Corollary 20.1.4,  $\mathbf{x} \in \text{conv}\{\mathbf{P}_\sigma \mathbf{y} : \sigma \in S_n\}$ , so there are weights  $t_\sigma \geq 0$  with  $\sum_\sigma t_\sigma = 1$  and  $\mathbf{x} = \sum_\sigma t_\sigma \mathbf{P}_\sigma \mathbf{y}$ . By the triangle inequality (N3) and absolute homogeneity (N2) of Definition 15.1.1, applied to this finite sum,

$$\Phi(\mathbf{x}) \leq \sum_\sigma t_\sigma \Phi(\mathbf{P}_\sigma \mathbf{y}) = \sum_\sigma t_\sigma \Phi(\mathbf{y}) = \Phi(\mathbf{y}),$$

where the middle equality is (G1).

- (b) By [Lemma 20.3.3](#) there is a  $\mathbf{z}$  with **non-negative entries** satisfying  $\mathbf{x} \leq \mathbf{z}$  entrywise and  $\mathbf{z} \prec \mathbf{y}$ . Since  $\mathbf{x}$  and  $\mathbf{z}$  have non-negative entries,  $|x_i| = x_i \leq z_i = |z_i|$  for every  $i$ , so [Proposition 20.3.1](#) gives  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{z})$ . By part (a),  $\Phi(\mathbf{z}) \leq \Phi(\mathbf{y})$ . This proves the theorem. ■

Part (a) needs no sign condition, and part (b) cannot do without one.

**Warning. Part (b) is false for vectors with negative entries.** Take  $\mathbf{x} = (0, -10)$  and  $\mathbf{y} = (0, 0)$  in  $\mathbb{R}^2$ . The decreasing rearrangements are  $(0, -10)$  and  $(0, 0)$ , and the running totals  $0, -10$  are at most  $0, 0$ , so  $\mathbf{x} \prec_w \mathbf{y}$ . But  $\Phi(\mathbf{x}) > 0 = \Phi(\mathbf{y})$  for every norm  $\Phi$ . The lemma is not at fault —  $\mathbf{z} = \mathbf{0}$  does satisfy  $\mathbf{x} \leq \mathbf{z}$  and  $\mathbf{z} \prec \mathbf{y}$ . What fails is the step  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{z})$ , because [Proposition 20.3.1](#) compares **absolute values**, and  $|-10| > |0|$ . Weak majorization lets a vector run far in the negative direction, and a norm measures that; only equality of totals, as in  $\prec$ , or non-negativity of the entries rules it out. This is why Section 4 applies the theorem to singular values and never to eigenvalues.

The converse of part (b) says that the family of symmetric gauges is strong enough to *detect* weak majorization, and it costs nothing: the Ky Fan gauges  $\Phi_k$  of [Definition 20.3.2](#) are themselves symmetric gauges ([Example \(The p-norms, the Ky Fan gauges, and size one\)](#) (b)), so the inequalities  $\Phi_k(\mathbf{x}) \leq \Phi_k(\mathbf{y})$  for every  $k$  are among those hypothesized, and for non-negative vectors they are the definition of  $\mathbf{x} \prec_w \mathbf{y}$ . Exercise C1 below states the two directions together, and Section 6 turns them into a theorem about matrices.

## 20.3.5 The dual gauge

Chapter 17 §04 gave every norm on  $\mathbb{R}^n$  a dual,  $\|\mathbf{y}\|_* = \max\{\langle \mathbf{x}, \mathbf{y} \rangle : \|\mathbf{x}\| \leq 1\}$  (Definition 17.4.4), and proved that taking the dual twice returns the norm (Theorem 17.4.8). Applied to a symmetric gauge, the construction stays inside the class.

### Definition 20.3.3 — Dual Gauge

Let  $\Phi$  be a symmetric gauge function on  $\mathbb{R}^n$ . Its **dual gauge** is the dual norm of



Definition 17.4.4,

$$\Phi^*(\mathbf{y}) = \max\{\langle \mathbf{x}, \mathbf{y} \rangle : \mathbf{x} \in \mathbb{R}^n, \Phi(\mathbf{x}) \leq 1\}.$$

### Theorem 20.3.5 — The Dual of a Symmetric Gauge

Let  $\Phi$  be a symmetric gauge function on  $\mathbb{R}^n$ . Then  $\Phi^*$  is a symmetric gauge function, and  $\Phi^{**} = \Phi$ .

*Idea.* Being a norm and being its own double dual are Chapter 17's, quoted. Only (G1) and (G2) need an argument, and each is the same one: the transformation can be moved off  $\mathbf{y}$  and onto  $\mathbf{x}$ , where it permutes the constraint set  $\{\Phi \leq 1\}$  onto itself.

*Proof.* That  $\Phi^*$  is a norm is Proposition 17.4.6 (a), and  $\Phi^{**} = \Phi$  is Theorem 17.4.8; both apply to any norm on  $\mathbb{R}^n$ .

(G1). Let  $\mathbf{P} = \mathbf{P}_\sigma$  be a permutation matrix. By Lemma 2.6.6 (b),  $\mathbf{P}^\top = \mathbf{P}^{-1}$ , so  $\langle \mathbf{x}, \mathbf{P}\mathbf{y} \rangle = \mathbf{x}^\top \mathbf{P}\mathbf{y} = (\mathbf{P}^\top \mathbf{x})^\top \mathbf{y} = \langle \mathbf{P}^\top \mathbf{x}, \mathbf{y} \rangle$ . As  $\mathbf{x}$  runs over  $\{\Phi \leq 1\}$ , so does  $\mathbf{x}' = \mathbf{P}^\top \mathbf{x}$ : indeed  $\Phi(\mathbf{P}^\top \mathbf{x}) = \Phi(\mathbf{x})$  by (G1) for  $\Phi$ , and  $\mathbf{x} \mapsto \mathbf{P}^\top \mathbf{x}$  is a bijection of  $\mathbb{R}^n$ . Hence the two maxima are over the same set of numbers, and  $\Phi^*(\mathbf{P}\mathbf{y}) = \Phi^*(\mathbf{y})$ .

(G2). Fix signs  $\varepsilon_i = \pm 1$  and write  $\mathbf{y}_\varepsilon = (\varepsilon_1 y_1, \dots, \varepsilon_n y_n)$ . Then  $\langle \mathbf{x}, \mathbf{y}_\varepsilon \rangle = \sum_i x_i \varepsilon_i y_i = \langle \mathbf{x}_\varepsilon, \mathbf{y} \rangle$ , and  $\mathbf{x} \mapsto \mathbf{x}_\varepsilon$  is a bijection of  $\mathbb{R}^n$  with  $\Phi(\mathbf{x}_\varepsilon) = \Phi(\mathbf{x})$  by (G2) for  $\Phi$ . So again the two maxima range over the same numbers, and  $\Phi^*(\mathbf{y}_\varepsilon) = \Phi^*(\mathbf{y})$ . This proves the theorem. ■

So duality is an involution of the set of symmetric gauge functions on  $\mathbb{R}^n$ . The examples are Chapter 17's: by Proposition 17.4.7 and Theorem 17.11.6,

$$(\|\cdot\|_p)^* = \|\cdot\|_q \quad \left( \frac{1}{p} + \frac{1}{q} = 1, \quad 1 \leq p \leq \infty \right),$$

so  $\|\cdot\|_1$  and  $\|\cdot\|_\infty$  are dual to each other and  $\|\cdot\|_2$  is self-dual. In the notation of Definition 20.3.2 that says  $\Phi_1^* = \Phi_\infty$  and  $\Phi_\infty^* = \Phi_1$ ; exercise C2 below computes  $\Phi_k^*$  for the  $k$  in between. Section 5 will find the matrix meaning of all of this: the dual of a unitarily invariant norm, taken with respect to the trace pairing, is the norm attached to the dual gauge.

## 20.3.6 Exercises

### 20.3.6.1 A. Check your understanding

#### Exercise A1.

- State Definition 20.3.1, naming both invariance clauses and the space the function is defined on.
- Explain why  $\Phi(\mathbf{x}) = \Phi(|\mathbf{x}|^\downarrow)$  for every symmetric gauge and every  $\mathbf{x}$ .
- True or false, with a reason: if  $\mathbf{x} \prec_w \mathbf{y}$  in  $\mathbb{R}^n$ , then  $\|\mathbf{x}\|_2 \leq \|\mathbf{y}\|_2$ .

- (d) Name the hypothesis in [Theorem 20.3.4](#) (b) that part (a) does without, and say where it is used in the proof.

*Solution.*

- (a) A symmetric gauge function on  $\mathbb{R}^n$  is a norm  $\Phi: \mathbb{R}^n \rightarrow \mathbb{R}$  satisfying (G1)  $\Phi(\mathbf{P}_\sigma \mathbf{x}) = \Phi(\mathbf{x})$  for every permutation matrix, and (G2)  $\Phi(\varepsilon_1 x_1, \dots, \varepsilon_n x_n) = \Phi(\mathbf{x})$  for all choices of signs  $\varepsilon_i = \pm 1$ .
- (b) This is [Eq. 20.3.1](#). Replacing each  $x_i$  by  $|x_i|$  is a choice of signs, so (G2) gives  $\Phi(\mathbf{x}) = \Phi(|\mathbf{x}|)$ ; sorting  $|\mathbf{x}|$  decreasingly is applying a permutation matrix, so (G1) gives  $\Phi(|\mathbf{x}|) = \Phi(|\mathbf{x}|^\downarrow)$ .
- (c) False in general. Take  $\mathbf{x} = (0, -10)$  and  $\mathbf{y} = (0, 0)$ : then  $\mathbf{x} \prec_w \mathbf{y}$  but  $\|\mathbf{x}\|_2 = 10 > 0$ . It is true when  $\mathbf{x}$  and  $\mathbf{y}$  have non-negative entries, by [Theorem 20.3.4](#) (b) with  $\Phi = \|\cdot\|_2$ .
- (d) Non-negativity of the entries. It is used to pass from  $\mathbf{x} \leq \mathbf{z}$  entrywise to  $|x_i| \leq |z_i|$ , which is what [Proposition 20.3.1](#) requires; without it,  $x_i \leq z_i$  says nothing about absolute values.

### 20.3.6.2 B. Practice

#### Exercise B1: Which of these are symmetric gauges.

Determine which of the following are symmetric gauge functions on  $\mathbb{R}^3$ . Justify your answer, naming the exact clause that fails when one does.

- (a)  $\Phi(\mathbf{x}) = |x_1| + |x_2| + |x_3|$ .
- (b)  $\Phi(\mathbf{x}) = |x_1|^\downarrow + |x_2|^\downarrow$ .
- (c)  $\Phi(\mathbf{x}) = (|x_1|^{1/2} + |x_2|^{1/2} + |x_3|^{1/2})^2$ .
- (d)  $\Phi(\mathbf{x}) = \max(|x_1|, |x_2|) + |x_3|$ .
- (e)  $\Phi(\mathbf{x}) = |x_1 + x_2 + x_3| + |x_1 - x_2| + |x_2 - x_3|$ .

*Solution.*

- (a) Yes: this is  $\|\cdot\|_1 = \Phi_3$ , [Example \(The p-norms, the Ky Fan gauges, and size one\)](#) (a).
- (b) Yes: this is the Ky Fan gauge  $\Phi_2$  of [Definition 20.3.2](#), checked in [Example \(The p-norms, the Ky Fan gauges, and size one\)](#) (b).
- (c) No. It is non-negative, absolutely homogeneous, permutation invariant and sign invariant, but (N3) fails: with  $\mathbf{x} = (1, 0, 0)$  and  $\mathbf{y} = (0, 1, 0)$  we get  $\Phi(\mathbf{x}) = \Phi(\mathbf{y}) = 1$  while  $\Phi(\mathbf{x} + \mathbf{y}) = (1 + 1)^2 = 4 > 2$ . It is not a norm, so it is not a symmetric gauge.
- (d) No. It is a norm — a sum of two seminorms, positive definite because each coordinate appears — and it is sign invariant, so (G2) holds. But (G1) fails:  $\Phi(1, 1, 0) = 1$  while  $\Phi(1, 0, 1) = 2$ .
- (e) No. It is a norm: it is non-negative and absolutely homogeneous, subadditive as a sum of three seminorms, and  $\Phi(\mathbf{x}) = 0$  forces  $x_1 = x_2 = x_3$  and  $|3x_1| = 0$ , hence  $\mathbf{x} = \mathbf{0}$ . Both invariance clauses fail. For (G1),  $\Phi(1, 1, 0) = 2 + 0 + 1 = 3$  while  $\Phi(1, 0, 1) = 2 + 1 + 1 = 4$ . For (G2),  $\Phi(1, 1, 1) = 3 + 0 + 0 = 3$  while

$$\Phi(1, -1, 1) = 1 + 2 + 2 = 5.$$

**Exercise B2: Filling from the bottom.**

Let  $\mathbf{x} = (6, 2, 2, 0)$  and  $\mathbf{y} = (7, 4, 3, 2)$  in  $\mathbb{R}^4$ .

- Verify that  $\mathbf{x} \prec_w \mathbf{y}$  and that  $\mathbf{x} \not\prec \mathbf{y}$ .
- Carry out the construction in the proof of [Lemma 20.3.3](#) to produce  $\mathbf{z}$  with  $\mathbf{x} \leq \mathbf{z}$  and  $\mathbf{z} \prec \mathbf{y}$ .
- Hence bound  $\|\mathbf{x}\|_2$  by  $\|\mathbf{y}\|_2$  without computing either, and then check the bound numerically.

*Solution.*

- Both are decreasing. Running totals: 6, 8, 10, 10 for  $\mathbf{x}$  and 7, 11, 14, 16 for  $\mathbf{y}$ . Each is at most the corresponding one, so  $\mathbf{x} \prec_w \mathbf{y}$ ; the totals 10 and 16 differ, so  $\mathbf{x} \not\prec \mathbf{y}$ .
- Here  $S = 16$  and  $n = 4$ , so  $c_0 = 16/4 = 4$ ,  $c_1 = (16 - 6)/3 = 10/3$ ,  $c_2 = (16 - 8)/2 = 4$ ,  $c_3 = 16 - 10 = 6$ . Test  $c_k \geq x_{k+1}$ :  $c_0 = 4 < 6 = x_1$ , so  $k = 0$  fails;  $c_1 = 10/3 \geq 2 = x_2$ , so  $j = 1$  and  $c = 10/3$ . Hence

$$\mathbf{z} = (6, \frac{10}{3}, \frac{10}{3}, \frac{10}{3}).$$

Check:  $\mathbf{x} \leq \mathbf{z}$  entrywise;  $\mathbf{z}$  is decreasing; running totals 6,  $\frac{28}{3}$ ,  $\frac{38}{3}$ , 16, and  $6 \leq 7$ ,  $\frac{28}{3} \approx 9.33 \leq 11$ ,  $\frac{38}{3} \approx 12.67 \leq 14$ ,  $16 = 16$ . So  $\mathbf{z} \prec \mathbf{y}$ .

- Both vectors have non-negative entries, so [Theorem 20.3.4](#) (b) with  $\Phi = \|\cdot\|_2$  gives  $\|\mathbf{x}\|_2 \leq \|\mathbf{y}\|_2$ . Numerically  $\|\mathbf{x}\|_2 = \sqrt{36 + 4 + 4} = \sqrt{44}$  and  $\|\mathbf{y}\|_2 = \sqrt{49 + 16 + 9 + 4} = \sqrt{78}$ , and  $44 \leq 78$ . The intermediate vector satisfies  $\|\mathbf{z}\|_2^2 = 36 + 3 \cdot \frac{100}{9} = \frac{208}{3} \approx 69.3$ , between the two, as the proof predicts.

**Exercise B3: Monotonicity used twice.**

Let  $\Phi$  be a symmetric gauge function on  $\mathbb{R}^3$  with  $\Phi(\mathbf{e}_1) = 1$ .

- Prove that  $1 \leq \Phi(1, 1, 0) \leq 2$ .
- Prove that  $\Phi(2, 1, 1) \leq \Phi(3, 1, 0)$ .
- Give a symmetric gauge for which the inequality in (b) is strict, and one for which it is an equality.

*Solution.*

- The Quick check following [Proposition 20.3.1](#) gives  $\Phi(\mathbf{e}_1)\|\mathbf{x}\|_\infty \leq \Phi(\mathbf{x}) \leq \Phi(\mathbf{e}_1)\|\mathbf{x}\|_1$ . With  $\mathbf{x} = (1, 1, 0)$  and  $\Phi(\mathbf{e}_1) = 1$  this reads  $1 \leq \Phi(1, 1, 0) \leq 2$ .
- Both vectors are decreasing with total 4, and the running totals 2, 3 of  $(2, 1, 1)$  are at most the running totals 3, 4 of  $(3, 1, 0)$ , so  $(2, 1, 1) \prec (3, 1, 0)$ . Now [Theorem 20.3.4](#) (a) applies.
- Strict for  $\Phi = \|\cdot\|_2$ :  $\sqrt{6} < \sqrt{10}$ . It is also strict for  $\Phi = \Phi_2$ , where the values are 3 and 4. Equality holds for  $\Phi = \|\cdot\|_1 = \Phi_3$ , where both values are 4: any two non-negative vectors with the same total have the same 1-norm, so a majorization

can never be detected by  $\Phi_3$  alone. This is the general shape of the equality case — the top gauge sees only the total, which a majorization preserves.

### 20.3.6.3 C. Going deeper

#### Exercise C1: Symmetric gauges detect weak majorization.

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  have non-negative entries.

- (a) Prove that  $\mathbf{x} \prec_w \mathbf{y}$  if and only if  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$  for **every** symmetric gauge function  $\Phi$  on  $\mathbb{R}^n$ .
- (b) Prove that  $\mathbf{x} \prec \mathbf{y}$  if and only if  $\mathbf{x} \prec_w \mathbf{y}$  and  $\sum_i x_i = \sum_i y_i$ .
- (c) Deduce that  $\mathbf{x} \prec \mathbf{y}$  if and only if  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$  for every symmetric gauge and  $\sum_i x_i = \sum_i y_i$ .

*Hint for (a): only finitely many gauges are needed.*

*Solution.*

- (a)  $(\Rightarrow)$  is [Theorem 20.3.4](#) (b).  $(\Leftarrow)$  Apply the hypothesis to the Ky Fan gauges  $\Phi_k$  of [Definition 20.3.2](#),  $k = 1, \dots, n$ . Since  $\mathbf{x}$  and  $\mathbf{y}$  have non-negative entries,  $|\mathbf{x}|^\downarrow = \mathbf{x}^\downarrow$  and likewise for  $\mathbf{y}$ , so  $\Phi_k(\mathbf{x}) = \sum_{i \leq k} x_i^\downarrow$  and  $\Phi_k(\mathbf{y}) = \sum_{i \leq k} y_i^\downarrow$ . The inequalities  $\Phi_k(\mathbf{x}) \leq \Phi_k(\mathbf{y})$  for  $k = 1, \dots, n$  are exactly (M1) extended to  $k = n$ , which is Definition 16.8.1's  $\mathbf{x} \prec_w \mathbf{y}$ .
- (b) This is a restatement of Definition 16.8.1:  $\mathbf{x} \prec \mathbf{y}$  means (M1) for  $k \leq n-1$  together with equality of totals, and  $\mathbf{x} \prec_w \mathbf{y}$  means (M1) for  $k \leq n$ . Given equality of totals, the case  $k = n$  of (M1) is automatic, so the two lists of conditions agree.
- (c) Combine (a) and (b):  $\mathbf{x} \prec \mathbf{y}$  holds exactly when  $\mathbf{x} \prec_w \mathbf{y}$  and the totals are equal, and by (a) the first of these is equivalent to  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$  for every symmetric gauge.

#### Exercise C2: The dual of a Ky Fan gauge.

Fix  $1 \leq k \leq n$  and let  $\Phi_k$  be the Ky Fan gauge of [Definition 20.3.2](#).

- (a) Prove that  $\Phi_k^*(\mathbf{y}) \geq \max(\|\mathbf{y}\|_\infty, \frac{1}{k}\|\mathbf{y}\|_1)$  by exhibiting two competitors.
- (b) Let  $\Phi_k(\mathbf{x}) \leq 1$  and let  $\mathbf{y}$  have non-negative, decreasing entries. Put  $t = |x|_k^\downarrow$  and split each  $|x_i|$  as  $(|x_i| - t)_+ + \min(|x_i|, t)$ . Prove that  $\sum_i (|x_i| - t)_+ \leq 1 - kt$  and that  $0 \leq t \leq 1/k$ .
- (c) Deduce  $\Phi_k^*(\mathbf{y}) = \max(\|\mathbf{y}\|_\infty, \frac{1}{k}\|\mathbf{y}\|_1)$ , and check the cases  $k = 1$  and  $k = n$  against Proposition 17.4.7.

*Solution.*

- (a) The vector  $\mathbf{e}_1$  has  $\Phi_k(\mathbf{e}_1) = 1$ ; replacing it by  $\pm \mathbf{e}_j$  with the sign of  $y_j$ , for a  $j$  where  $|y_j|$  is largest, gives  $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{y}\|_\infty$ . The vector  $\mathbf{x}$  with  $x_i = \pm 1/k$ , the sign of  $y_i$ , has  $\Phi_k(\mathbf{x}) = k \cdot \frac{1}{k} = 1$  and  $\langle \mathbf{x}, \mathbf{y} \rangle = \frac{1}{k}\|\mathbf{y}\|_1$ . So  $\Phi_k^*(\mathbf{y})$  is at least both numbers.

- (b) Since  $|x|^\downarrow$  is decreasing,  $(|x|_i^\downarrow - t)_+ = 0$  for  $i > k$ , so

$$\sum_i (|x_i| - t)_+ = \sum_{i \leq k} (|x|_i^\downarrow - t) = \Phi_k(\mathbf{x}) - kt \leq 1 - kt.$$

Also  $kt \leq \sum_{i \leq k} |x|_i^\downarrow = \Phi_k(\mathbf{x}) \leq 1$ , so  $0 \leq t \leq 1/k$ .

- (c) By [Theorem 20.3.5](#),  $\Phi_k^*$  is a symmetric gauge, so its value at  $\mathbf{y}$  depends only on  $|\mathbf{y}|^\downarrow$ ; and replacing  $\mathbf{x}$  by  $|\mathbf{x}|$  leaves  $\Phi_k(\mathbf{x})$  unchanged and does not decrease  $\langle \mathbf{x}, \mathbf{y} \rangle$  when  $\mathbf{y} \geq \mathbf{0}$ . So we may assume  $\mathbf{y}$  has non-negative decreasing entries and  $\mathbf{x} \geq \mathbf{0}$ . With  $t$  as in (b), and using  $y_i \leq y_1 = \|\mathbf{y}\|_\infty$  in the first sum and  $\min(x_i, t) \leq t$  in the second,

$$\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{y}\|_\infty(1 - kt) + t\|\mathbf{y}\|_1 = \|\mathbf{y}\|_\infty + t(\|\mathbf{y}\|_1 - k\|\mathbf{y}\|_\infty).$$

If  $\|\mathbf{y}\|_1 \leq k\|\mathbf{y}\|_\infty$ , the bracket is at most 0 and  $t \geq 0$ , so the right side is at most  $\|\mathbf{y}\|_\infty$ . If  $\|\mathbf{y}\|_1 > k\|\mathbf{y}\|_\infty$ , the right side increases with  $t$ , and  $t \leq 1/k$  gives the bound  $\|\mathbf{y}\|_\infty + \frac{1}{k}\|\mathbf{y}\|_1 - \|\mathbf{y}\|_\infty = \frac{1}{k}\|\mathbf{y}\|_1$ . In both cases  $\langle \mathbf{x}, \mathbf{y} \rangle \leq \max(\|\mathbf{y}\|_\infty, \frac{1}{k}\|\mathbf{y}\|_1)$ , which with (a) gives equality.

For  $k = 1$ ,  $\Phi_1 = \|\cdot\|_\infty$  and the formula gives  $\max(\|\mathbf{y}\|_\infty, \|\mathbf{y}\|_1) = \|\mathbf{y}\|_1$ , matching [Proposition 17.4.7 \(b\)](#). For  $k = n$ ,  $\Phi_n = \|\cdot\|_1$  and the formula gives  $\max(\|\mathbf{y}\|_\infty, \frac{1}{n}\|\mathbf{y}\|_1) = \|\mathbf{y}\|_\infty$ , matching part (a) of the same proposition.

### Exercise C3: Where each hypothesis is used.

- Determine whether there is a norm on  $\mathbb{R}^2$  that is permutation invariant and satisfies  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$  whenever  $|x_i| \leq |y_i|$  for all  $i$ , but is **not** a symmetric gauge function. Justify your answer.
- [Theorem 20.3.4 \(b\)](#) needs non-negative entries, while [Lemma 20.3.3](#) does not. Locate the exact step of the proof of the theorem that uses the hypothesis, and explain why  $\mathbf{x} = (0, -10) \prec_w (0, 0) = \mathbf{y}$  breaks it even though the lemma supplies a perfectly good  $\mathbf{z}$ .
- Let  $\mathbf{x} \prec_w \mathbf{y}$  with  $\mathbf{x} = \mathbf{x}^\downarrow$ , and let  $\mathbf{z}$  be the vector built in the proof of [Lemma 20.3.3](#). Prove that  $\mathbf{z} \prec \mathbf{z}'$  for **every**  $\mathbf{z}'$  with  $\mathbf{x} \leq \mathbf{z}'$  and  $\sum_i z'_i = \sum_i y_i$ , and deduce that  $\sum_i \phi(z_i) \leq \sum_i \phi(z'_i)$  for every convex  $\phi$ . *Hint: compare top partial sums; Theorem 20.2.1 does the last step.*

*Solution.*

- There is none: monotonicity in the stated sense already forces (G2). If  $|x_i| \leq |y_i|$  implies  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{y})$ , then for any signs the vectors  $\mathbf{x}$  and  $(\varepsilon_1 x_1, \dots, \varepsilon_n x_n)$  dominate each other entrywise in absolute value, so their  $\Phi$ -values are equal, which is (G2). With permutation invariance this is [Definition 20.3.1](#). So the conclusion of [Proposition 20.3.1](#) is not merely implied by (G2): for any norm the two are equivalent.
- The step is  $\Phi(\mathbf{x}) \leq \Phi(\mathbf{z})$ , which invokes [Proposition 20.3.1](#) and therefore needs  $|x_i| \leq |z_i|$ , not  $x_i \leq z_i$ . For  $\mathbf{x} = (0, -10)$  and  $\mathbf{y} = (0, 0)$  the lemma does supply a  $\mathbf{z}$ , namely  $\mathbf{z} = \mathbf{0}$ : indeed  $\mathbf{z} \prec \mathbf{y}$  and  $\mathbf{x} \leq \mathbf{z}$ . But  $|x_2| = 10 > 0 = |z_2|$ , so [Proposition 20.3.1](#) does not apply, and it must not:  $\Phi(\mathbf{x}) > 0 = \Phi(\mathbf{z})$  for every norm. When the entries are non-negative,  $x_i \leq z_i$  and  $|x_i| \leq |z_i|$  say the same

thing, and the gap closes.

- (c) Write  $S = \sum_i y_i$  and recall  $\mathbf{z} = (x_1, \dots, x_j, c, \dots, c)$ , decreasing, with  $\sum_i z_i = S$ . Let  $\mathbf{z}'$  satisfy  $\mathbf{x} \leq \mathbf{z}'$  and  $\sum_i z'_i = S$ , and write  $M_1 \geq \dots \geq M_n$  for its entries in decreasing order. Since the totals agree, (M2) holds, and it remains to check (M1) for  $1 \leq k \leq n-1$ .

*Case  $k \leq j$ .* Then  $\sum_{i \leq k} z_i = \sum_{i \leq k} x_i \leq \sum_{i \leq k} z'_i \leq \sum_{i \leq k} M_i$ , the first inequality from  $\mathbf{x} \leq \mathbf{z}'$  and the second because no  $k$  entries of  $\mathbf{z}'$  add up to more than its  $k$  largest.

*Case  $k > j$ .* Here  $\sum_{i \leq k} z_i = S - (n-k)c$ , so the claim is  $\sum_{i > k} M_i \leq (n-k)c$ . Suppose not. Then the average of  $M_{k+1}, \dots, M_n$  exceeds  $c$ , and  $M_{k+1}$ , being the largest of them, satisfies  $M_{k+1} > c$ ; hence  $M_i > c$  for every  $i \leq k+1$ . Also  $\sum_{i \leq j} M_i \geq \sum_{i \leq j} x_i$ , by the argument of the previous case. Therefore

$$\begin{aligned} S = \sum_i M_i &= \sum_{i \leq j} M_i + \sum_{i=j+1}^k M_i + \sum_{i > k} M_i \\ &> \sum_{i \leq j} x_i + (k-j)c + (n-k)c = S, \end{aligned}$$

the last equality being  $\sum_{i \leq j} x_i + (n-j)c = S$ , which defined  $c$ . This contradiction proves the case.

So  $\mathbf{z} \prec \mathbf{z}'$ , and Theorem 20.2.1 gives  $\sum_i \phi(z_i) \leq \sum_i \phi(z'_i)$  for every convex  $\phi$ . In words: among all admissible ways of raising  $\mathbf{x}$  to the required total, the flat filling is the most even one.